

REZA SHIRKAVAND

Department of Computer Science, University of Maryland, College Park, MD, USA

[Email](#), [Website](#), [Google Scholar](#), [Github](#)

RESEARCH INTERESTS

Adaptive AI

EDUCATION

Sharif University of Technology *2020*

B.Sc. in Computer Engineering

University of Pittsburgh *2022 - 2023*

M.Sc. in Electrical & Computer Engineering

University of Maryland *2023 - Present*

Ph.D. in Computer Science

SELECTED PUBLICATIONS

Catalog-Native LLM: Speaking Item-ID Dialect with Less Entanglement for Recommendation

R. Shirkavand, X. Wei, C. Wang, Z. Hui, H. Huang, M. Gong

[ICLR 2026](#)

Cost-Aware Contrastive Routing for LLMs

R. Shirkavand, S. Gao, P. Yu, H. Huang

[NeurIPS 2025 \(Spotlight\)](#)

Bilevel ZOFO: Bridging Parameter-Efficient and Zeroth-Order Techniques for Efficient LLM Fine-Tuning and Meta-Training

R. Shirkavand, P. Yu, Q. He, H. Huang

[NeurIPS 2025](#)

Efficient Fine-Tuning and Concept Suppression for Pruned Diffusion Models

R. Shirkavand, P. Yu, S. Gao, G. Somepalli, T. Goldstein, H. Huang

[CVPR 2025](#)

Not All Prompts Are Made Equal: Prompt-based Pruning of Text-to-Image Diffusion Models

R. Shirkavand*, A. Ganjdanesh*, S. Gao, H. Huang

[ICLR 2025](#)

Flash-BoN: Instant Drafts for Inference-Time Scaling in Diffusion Models

R. Rawal, **R. Shirkavand**, S. Paul, Y. Wen, H. Huang, Y. Chen, G. Somepalli, T. Goldstein

[ECCV 2026](#)

ToMoE: Converting Dense Large Language Models to Mixture-of-Experts through Dynamic Structural Pruning

S. Gao, T. Hua, **R. Shirkavand**, C. Lin, et. al

[TMLR @ICML 2026](#)

ARGUS: Hallucination and Omission Evaluation in Video-LLMs
*R. Rawal, **R. Shirkavand**, H. Huang, G. Somepalli, T. Goldstein*
ICCV 2025

From Pixels to Prose: A Large Dataset of Dense Image Captions
*V. Singla, K. Yue, **R. Shirkavand**, S. Paul, et al.*
Preprint

WORK EXPERIENCE

PhD AI Research Intern Jun 2026 - Aug 2026
Two Sigma

Worked on pre-training of Financial Foundation Models.

PhD AI Research Intern Jun 2025 - Dec 2025
Roblox

Worked on Mixture of Experts for Recommendation and Generation.

TEACHING EXPERIENCE

Co-Instructor, Advanced Machine Learning Topics *Spring 2024,2025,2026*
Presented an overview of the transformer architecture and its applications.

Teaching Assistant, Systems & Projects Engineering *Summer 2022*
Taught essential topics in systems engineering and project management.

Teaching Assistant, Algorithmic Thinking *Spring 2022*
Delivered lectures and supported students in developing the theoretical and practical skills necessary for designing algorithms.

Teaching Assistant, Machine Learning *Fall 2019*
Supported students in gaining theoretical and practical knowledge in machine learning and statistical pattern recognition.

HONORS AND AWARDS

Nationwide University Entrance Exam - Mathematics *2015*
Ranked 14th among 181000 participants

Nationwide University Entrance Exam - Foreign Languages *2015*
Ranked 15th among 7000 participants

SKILLS

Programming Languages Python, Java, C++, Matlab

Libraries Pytorch, Tensorflow